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EditAI’s : An LLM-Based System for Public Procurement Tender Analysis

SÃO PAULO

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EditAI’s: An LLM-Based System for Public Procurement Tender Analysis

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OVERVIEW

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This work presents the development of an integrated system based on large language models (LLMs) to support the analysis of public procurement tender documents. The system allows users to upload tender documents and automatically performs the extraction and structuring of technical and financial requirements, as well as the cross-referencing of this information with the company's certificates, aiming to verify compliance with tender requirements. The adopted methodology involves the use of natural language processing techniques, a service-oriented architecture, and integration with corporate document repositories. The results demonstrate significant efficiency gains in opportunity analysis, a reduction in the time spent by sales and pre-sales teams, and increased agility in the decision-making process. It is concluded that the proposed solution contributes to enhancing the company's competitiveness in the public sector market by optimizing the tender participation process.

Keywords: Applied artificial intelligence; public procurement tenders; LLM summarization; language models; AI-driven process improvement.

ABSTRACT

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This work presents the development of an integrated system based on large language models (LLMs) to support the analysis of public procurement tender documents. The system allows users to upload tender documents and automatically performs the extraction and structuring of technical and financial requirements, as well as the cross-referencing of this information with the company's certificates, aiming to verify compliance with tender requirements. The adopted methodology involves the use of natural language processing techniques, a service-oriented architecture, and integration with corporate document repositories. The results demonstrate significant efficiency gains in opportunity analysis, a reduction in the time spent by sales and pre-sales teams, and increased agility in the decision-making process. It is concluded that the proposed solution contributes to enhancing the company's competitiveness in the public sector market by optimizing the tender participation process.

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1. Introduction

The analysis of public procurement tender documents represents a significant challenge for consulting companies operating in the public sector due to the high complexity, length, and diversity of technical and financial requirements contained in such documents. In many cases, tender documents may exceed several hundred pages, making manual interpretation a time-consuming, costly, and error-prone process, which directly impacts companies' ability to identify viable business opportunities.

Within the corporate context of ATOS, a consulting company that serves both public and private clients, this challenge becomes even more relevant, as sales and pre-sales teams are responsible for analyzing a large volume of public tenders, many of which do not ultimately result in awarded contracts. Delays in the analysis of these documents may lead to the loss of strategic opportunities, with significant financial impacts and constraints on the expansion of the company's presence in the public sector market.

In light of this scenario, there is a clear need for technological solutions capable of accelerating and optimizing the interpretation of tender documents, reducing the time devoted to analysis and increasing efficiency in the identification of genuine opportunities for participation in procurement processes. In this context, Artificial Intelligence, particularly large language models (LLMs), emerges as a promising approach for automating tasks related to the reading, extraction, and structuring of complex textual information.

Therefore, this work aims to develop an integrated system based on LLMs that enables the upload of tender documents and automatically extracts technical and financial requirements, as well as cross-references this information with the company's certificates in order to verify compliance with tender requirements. The proposed solution seeks to support sales and pre-sales teams by enhancing efficiency in opportunity analysis, optimizing time allocation, and increasing agility in the decision-making process, thereby contributing to improved competitiveness in the public sector market.

1.1. Partner Company Context:

The strategic relevance of this project lies in the need to increase efficiency and competitiveness in the analysis of public procurement tenders, a critical activity for consulting companies operating in the public sector. In 2024, the pre-sales team was involved in the analysis of approximately 50 public procurement projects, representing a total opportunity value of around R$ 500 million. From this volume, the company effectively participated in nine tenders, totaling approximately R$ 103 million, and successfully secured three projects, resulting in R$ 43 million in awarded contracts.

Despite these results, the current manual and time-intensive process of interpreting tender documents limits the company's ability to respond quickly to opportunities and allocate resources effectively, contributing to missed bids and reduced conversion rates throughout the qualification process. Given the high volume and value of opportunities handled by the pre-sales team, inefficiencies at this stage have a direct and significant impact on business outcomes.

By leveraging large language models to automate the extraction and analysis of technical and financial requirements, the proposed solution enables sales and pre-sales teams to focus on higher-value activities, such as strategic decision-making and client engagement. This directly contributes to reducing operational costs, improving response times, and increasing the likelihood of participating in tenders that are better aligned with the company's technical certifications and capabilities.

Furthermore, the project supports the company's long-term strategy of adopting artificial intelligence to optimize internal processes and enhance data-driven decision-making. By systematizing knowledge extracted from tender documents and aligning it with corporate certifications, the solution strengthens the company's positioning in the public sector market and promotes sustainable growth through more informed and agile participation in procurement processes.

1.2. Problem Definition (Corporate Pain Point):

The current process of analyzing public procurement tender documents within the company is predominantly manual and highly dependent on the availability and expertise of sales and pre-sales professionals. Each tender document must be read in full to identify technical, financial, and compliance requirements, a task that becomes increasingly complex due to the length and heterogeneity of these documents, which may exceed several hundred pages.

As a result, the process is time-consuming and inefficient, often requiring multiple working hours or even several days per tender, depending on its complexity. This manual approach also increases the likelihood of human error, such as overlooking critical requirements or misinterpreting technical specifications, which may lead to the disqualification of bids or the pursuit of opportunities that are not aligned with the company's certifications and capabilities.

To better understand the magnitude of this issue, a structured survey was conducted with pre-sales managers, revealing that the average time required to analyze a single tender document is approximately 34 hours. This baseline metric highlights a significant operational bottleneck, as it limits the number of opportunities that can be assessed within a given timeframe.

The absence of automated filtering and information extraction mechanisms contributes to elevated operational costs and low conversion rates, since a substantial portion of the analyzed tenders does not progress to formal participation. Additionally, the lack of structured data extracted from tender documents hinders historical analysis and continuous improvement of the bidding process.

1.3. Proposed Solution and Expected Contribution:

The proposed solution consists of the development of an integrated computational system based on large language models (LLMs) designed to support the automated analysis of public procurement tender documents. The system enables users to upload tender documents and automatically performs the extraction, structuring, and summarization of technical and financial requirements. In addition, it cross-references these requirements with the company's certifications and qualifications, allowing for a faster and more reliable assessment of eligibility for participation in each tender.

By automating tasks that are currently performed manually, the solution aims to significantly reduce the time required for tender analysis and minimize the risk of human error. Based on the identified baseline of approximately 34 working hours per tender, the project seeks to achieve a substantial reduction in analysis time, increasing the overall productivity of sales and pre-sales teams and enabling a larger number of opportunities to be evaluated within the same timeframe.

From a business perspective, the expected contribution extends beyond operational efficiency. In the previous year, the company successfully secured three public procurement projects, totaling approximately 43 million in revenue. Increasing the number of awarded projects by even a single additional tender represents a meaningful financial gain. By improving the speed and quality of opportunity assessment, the proposed solution aims to increase the likelihood of identifying high-potential tenders and, consequently, contribute to higher conversion rates and revenue growth.

Therefore, the expected contribution of this project is twofold: operational, through measurable gains in productivity and time reduction in tender analysis, and strategic, by supporting more informed decision-making and enhancing the company's competitiveness in the public sector market.

1.4. Business Objectives:

The business objectives of this project are focused on improving operational efficiency, increasing revenue potential, and strengthening the company's positioning in the public sector market. The expected results include:

- Reduce tender analysis time, enabling sales and pre-sales teams to evaluate a greater number of opportunities within the same operational capacity.

- Increase team productivity by reallocating effort from manual document analysis to higher-value activities such as strategic decision-making and bid qualification.

- Improve opportunity selection quality, reducing the pursuit of tenders that do not align with the company's technical certifications and capabilities.

- Support revenue growth, by increasing the likelihood of securing additional public procurement projects, where even a single additional awarded tender represents a significant financial gain.

- Enhance competitiveness in the public sector, through faster response times, better-informed decisions, and more consistent participation in high-potential tenders.

- Promote data-driven decision-making, by enabling structured data extraction from tender documents and facilitating historical analysis of bidding performance.

1.5. Structure of the thesis/dissertation:

This thesis is structured as follows. Chapter 1 introduces the project, presenting its context, motivation, problem definition, and business and technical objectives. Chapter 2 discusses the theoretical background related to artificial intelligence, large language models, and natural language processing, emphasizing their application to the analysis of public procurement tender documents. Chapter 3 details the proposed computational solution, describing the system architecture, design decisions, and methodologies adopted. Chapter 4 presents the implementation and evaluation of the solution, including the analysis of productivity gains and business impact. Finally, Chapter 5 concludes the thesis by summarizing the main contributions, identifying limitations, and outlining directions for future work.

2 Solution Development

2.1 Applied Rationale

The choice of large language models (LLMs) as the core technology of the proposed solution is justified by the unstructured and highly complex nature of public procurement tender documents. These documents contain heterogeneous information, including legal, technical, and financial requirements, often expressed in natural language and distributed across hundreds of pages. LLMs are particularly well suited to this context due to their ability to process large volumes of text, capture semantic relationships, and perform tasks such as information extraction, summarization, and classification with a high degree of flexibility.

However, relying solely on a standalone LLM may lead to limitations related to factual accuracy, contextual consistency, and traceability of outputs. To address these challenges, the solution adopts a Retrieval-Augmented Generation (RAG) approach, which enables the model to retrieve relevant excerpts from tender documents and corporate certifications before generating responses. This method increases reliability and reduces the risk of hallucinations, which is a critical requirement in a business context where incorrect interpretations may result in disqualification from tenders or financial losses.

From an architectural perspective, the solution is designed following a modular and service-oriented approach, allowing independent evolution of components such as document ingestion, retrieval, model inference, and user interface. This design supports scalability, as it enables the system to handle fluctuations in demand, and facilitates maintainability and future extensions without impacting core functionalities.

The selected technologies and methods are aligned with the business requirements identified for the project, particularly the need to significantly reduce analysis time, minimize operational costs, and improve decision-making quality. By automating labor-intensive tasks while preserving human oversight for strategic decisions, the proposed solution achieves an effective balance between efficiency, reliability, and business control.

2.1.1 **Business Area Rationale:**

In consulting companies operating in the public sector, the pre-sales function plays a strategic role in the identification, qualification, and prioritization of business opportunities. Pre-sales teams are responsible for analyzing public procurement tenders, assessing technical and financial requirements, and supporting Go / No-Go decisions that determine whether the company should invest resources in participating in a given bid. This activity directly impacts revenue generation, competitiveness, and the efficient allocation of specialized human resources.

The analysis of public tenders involves complex bid management processes, including the interpretation of extensive technical documentation, evaluation of compliance requirements, risk assessment, and verification of the company's delivery capabilities. These activities are typically performed by highly specialized professionals with a high cost per hour, which makes the efficiency of the analysis process a critical factor for business performance. When conducted manually, tender analysis becomes time-consuming and limits the number of opportunities that can be evaluated within tight procurement deadlines.

From a business perspective, opportunity cost is a central concern in the pre-sales domain. Excessive time spent analyzing tenders that ultimately prove to be non-viable reduces the team's capacity to focus on opportunities with higher strategic alignment and success probability. In the context of public procurement, this challenge is intensified by the regulated nature of the process, the high volume and complexity of tender documents, and the short response windows imposed by public entities.

Market best practices indicate a growing adoption of automation and artificial intelligence to support pre-sales and bid management activities. Consulting firms increasingly rely on document intelligence, natural language processing, and AI-driven decision support tools to accelerate the extraction of technical and financial requirements, improve compliance analysis, and enable faster Go / No-Go decisions. These practices aim to reduce manual effort, minimize errors, and enhance the

strategic focus of pre-sales teams by allowing professionals to concentrate on higher-value tasks such as opportunity assessment and client strategy.

In this context, the proposed solution aligns with established market trends by leveraging large language models to automate critical stages of tender analysis. By systematizing the interpretation of complex procurement documents and supporting compliance verification, the solution addresses key pain points in the pre-sales process and reinforces best practices adopted in the consulting sector for public procurement.

2.1.2 **Technological rationale for the solution:**

The proposed solution is grounded in contemporary concepts and technologies that have emerged as market best practices for large-scale document analysis and intelligent automation. Given the complexity, volume, and unstructured nature of public procurement tender documents, the adoption of advanced natural language processing techniques and scalable cloud infrastructure is essential to ensure efficiency, reliability, and adaptability to business demands.

Large Language Models (LLMs) were selected as the core technology due to their proven capability to understand, summarize, and extract structured information from complex textual content. LLMs enable the automated identification of technical and financial requirements, significantly reducing manual interpretation effort and supporting faster and more consistent analysis. However, given the criticality of accuracy and traceability in public procurement contexts, LLMs were complemented with Retrieval-Augmented Generation (RAG). This approach allows the model to ground its responses in retrieved, relevant document segments, improving reliability, reducing hallucinations, and ensuring that outputs are based on verifiable source information.

To support efficient information retrieval, a vector database was employed to store document embeddings generated from tender documents and corporate certifications. Vector-based semantic search enables the system to retrieve

contextually relevant information with high precision, which is particularly important when dealing with large documents that may exceed hundreds of pages. This architecture enhances scalability and ensures consistent performance even as the volume of processed documents increases.

From an infrastructure perspective, the solution was deployed on Google Cloud Platform (GCP), leveraging a serverless architecture through Cloud Run. This choice aligns with modern cloud-native principles, offering automatic scalability, high availability, and cost efficiency by provisioning resources on demand. The use of Cloud Run allows the system to handle variable workloads, which is characteristic of tender analysis processes that experience peaks near bid deadlines.

The selection of GCP was further justified by the strategic partnership between ATOS and Google, which facilitates access to cloud services, technical support, and best practices aligned with the company's existing infrastructure and governance standards. This alignment reduces operational friction, accelerates deployment, and ensures compatibility with enterprise-level security and compliance requirements.

Overall, the combination of LLMs, vector databases, and RAG, deployed within a serverless cloud environment, represents a state-of-the-art technological stack for intelligent document analysis. This architecture not only addresses current operational challenges in pre-sales tender analysis but also provides a flexible and extensible foundation for future enhancements and broader adoption of AI-driven solutions within the organization.

2.1.3 **Fundamentals of Management and Development Methods:**

The development of the proposed solution was guided by agile principles, with a particular focus on the Scrum framework, which is widely adopted in corporate environments to manage complex and uncertain projects. Scrum was selected due to its emphasis on iterative development, frequent feedback, and continuous value delivery, all of which are essential in projects involving emerging technologies such as artificial intelligence.

The project was organized into short development cycles, or sprints, with a duration of two weeks. Each sprint aimed to deliver a functional increment of the solution, prioritized according to business value for the partner company. This approach enabled early validation of features with stakeholders from the pre-sales area, ensuring that the system evolved in alignment with real operational needs and reducing the risk of developing functionalities with limited practical impact.

Throughout the project, the backlog was continuously refined to reflect feedback received from users and changes in business priorities. Sprint planning, review, and retrospective ceremonies were used to promote transparency, align expectations, and foster continuous improvement in both the technical solution and the development process. This iterative workflow allowed for incremental enhancements in model performance, user interface usability, and system integration.

From a management perspective, the Scrum-based approach was aligned with principles from the Project Management Body of Knowledge (PMBOK), particularly in areas such as scope management, stakeholder engagement, risk management, and quality control. This combination ensured that while development remained flexible and adaptive, the project maintained structured governance and clear objectives within the corporate environment.

By adopting agile and Scrum practices, the project was able to focus on delivering tangible value every two weeks, optimize resource allocation, and maintain close collaboration with the partner company. This methodology proved effective in supporting rapid experimentation, accommodating evolving requirements, and delivering a solution that meets both technical and business expectations.

2.2 Specification and Development :

This section presents the technical specifications of the proposed solution, ensuring alignment with the standards, operational needs, and strategic objectives of the partner company. The specifications were defined based on interactions with stakeholders from the pre-sales area and reflect both business requirements and technical constraints associated with the analysis of public procurement tenders.

To ensure a structured and comprehensive definition of system capabilities, the requirements were gathered and categorized into functional and non-functional requirements. This distinction allows the solution to address not only the core functionalities expected by users, but also critical quality attributes such as performance, scalability, security, and usability, which are essential in a corporate environment.

Additionally, requirements were classified as mandatory or desirable. Mandatory requirements represent the minimum viable functionality required for the initial delivery of the solution, while desirable requirements correspond to value-added features planned for future iterations, aligned with the agile development approach adopted in the project.

2.2.1 **Requirements and Specifications:**

To ensure the project fully meets the client's needs, critical requirements were identified during a scoping interview aimed at refining the project's understanding. These requirements were categorized into two main groups: *functional* and *non-functional*. This division reflects the project's complexity, which demands both the implementation of core functionalities aligned with minimum standards and adherence to rigorous performance metrics. **Functional Requirements (FRs)** specify the actions and operations the system must perform, while **Non-Functional Requirements (NFRs)** define measurable criteria for optimization, security, scalability, and other quality attributes. Additionally, each requirement was classified as *"mandatory"* (essential for the initial delivery) or *"desirable"* (value-added for future iterations), enabling the team to strategically prioritize implementation in alignment with the client's objectives.

Table 1 - Functional Requirements (FRs)

ID	Title	Description	Category
FR1	Bid Document Processing and Summarization	The system must process bid documents by reading, analyzing, and summarizing their content to automatically extract technical and functional requirements.	Mandatory
FR2	Interactive Results	The system must dynamically generate a concise summary of key bid document insights and display them in a user-friendly prompt.	Mandatory

	Overview & Export	Additionally, it must enable on-demand export of the summarized results as a PDF for offline access.	
FR3	Bid-Compliance Certification Matching	The system must automatically cross-reference bid requirements with the company's existing certifications to identify and suggest relevant documentation that validates the company's capability to fulfill the bid's specific demands. This feature will leverage a vector database to intelligently categorize project types and align them with pre-approved certifications, ensuring contextual relevance.	Desirable
FR4	Comprehensive Usage History Archiving	The system must securely store all user requests, feedback, and system-generated compliance matches in a structured, searchable repository. This historical data will be aggregated into a centralized data lake, enabling long-term strategic analytics, trend identification, and operational auditability to support data-driven decision-making.	Desirable

FR5	Data Lake Analytics Dashboard	The system must include an interactive analytics interface that enables users to explore, filter, and visualize data from the centralized data lake. This dashboard will display user activity metrics, system performance trends, and compliance match insights through dynamic charts, tables, and customizable filters, empowering stakeholders to derive actionable intelligence and monitor long-term operational efficiency.	Desirable

Table 2 - Non-Functional Requirements (NFRs)

ID	Title	Description	Category
NFR1	Bid Document Processing Performance	system shall process bid documents (up to 10 MB in size) with an average processing time of ≤5 minutes, measured from upload completion to generation of the final output (technical/financial requirements summary and compliance matching).	Mandatory
NFR2	Scalability	The system must be able to process multiple bid documents simultaneously by distributing the workload.	Desirable
NFR3	Security	Data must be securely stored, ensuring file encryption and authenticated access via login.	Desirable
NFR4	Usability	The interface must be intuitive, with a minimal learning curve for sales and pre-sales users.	Desirable
NFR5	Logging	A logging mechanism must be in place to track failures, system usage, and user activities, facilitating audits and troubleshooting.	Desirable

2.2.2 **Architecture and Technology:**

The proposed solution adopts a cloud-native, client-server architecture designed to support scalability, reliability, and seamless integration with the partner company's IT ecosystem. The system is structured around modular components that separate user interaction, application logic, and data processing, enabling flexibility and ease of maintenance.

From an architectural perspective, the solution follows a service-oriented approach deployed in a cloud environment. The frontend acts as the client layer, providing an intuitive web interface that allows users from the sales and pre-sales teams to upload tender documents, visualize extracted requirements, and export analysis results. This client layer communicates with the backend through secure RESTful APIs.

The backend is responsible for orchestrating document processing workflows, including text extraction, embedding generation, semantic retrieval using a vector database, and interaction with large language models through a Retrieval-Augmented Generation (RAG) pipeline. This design ensures that the analysis of tender documents is both scalable and traceable, as responses are grounded in retrieved document segments rather than generated in isolation.

The entire system is deployed on Google Cloud Platform using Cloud Run, leveraging a serverless architecture that automatically scales according to demand. This approach is particularly suitable for the pre-sales context, where workload peaks are common near bid submission deadlines. By adopting a serverless model, the solution optimizes infrastructure costs while maintaining high availability and performance.

Integration with the company's IT ecosystem is achieved through standardized interfaces and cloud-native services. The solution can be connected to existing document repositories, authentication mechanisms, and corporate data sources, such as certification and compliance databases, without disrupting current workflows.

The use of managed cloud services aligns with the company's governance, security, and compliance requirements, facilitated by the existing strategic partnership between ATOS and Google.

Overall, the proposed architecture provides a robust and extensible foundation that supports current operational needs while enabling future enhancements, such as integration with CRM systems, business intelligence platforms, and advanced analytics tools. This architectural approach ensures that the solution can evolve alongside the company's digital transformation initiatives.

2.2.3 **Development and Implementation (MVP):**

The development and implementation of the proposed solution followed an agile methodology, primarily based on the Scrum framework, which is well suited for projects involving innovation, uncertainty, and continuous stakeholder interaction. Scrum was adopted to enable incremental deliveries, frequent validation, and continuous alignment with the business needs of the partner company.

The project was structured into two-week sprints, allowing the team to deliver functional increments of the solution at the end of each iteration. This approach ensured that value was continuously generated for the pre-sales team, while enabling rapid feedback and iterative refinement of both technical and functional aspects of the system. Sprint ceremonies, including planning, review, and retrospective meetings, were conducted regularly to maintain transparency, assess progress, and identify improvement opportunities.

Project Roadmap and Macro Schedule

To ensure efficient organization and execution, the project was divided into modular phases, each addressing a critical component of the solution. The project started on February 4th, 2025, and was planned to span approximately 40 weeks, comprising 20 sprints. Each module was developed over five sprints, providing a clear structure and measurable milestones.

The first module focused on the development and use of Large Language Models (LLMs), including requirements analysis, initial model implementation, model selection, fine-tuning strategies, and validation based on early feedback. This phase established the foundation for automated extraction and interpretation of tender documents.

The second module addressed frontend development, covering interface requirements definition, UI/UX design, implementation of document upload functionality, and integration with backend services. Usability testing and iterative refinements were conducted to ensure alignment with the workflows of sales and pre-sales users.

The third module concentrated on the implementation of the Retrieval-Augmented Generation (RAG) pipeline. This phase involved defining data retrieval strategies, developing the RAG workflow, integrating it with the LLM, and validating the reliability and accuracy of grounded responses through systematic testing.

The fourth module focused on system scalability and robustness. Activities included infrastructure optimization, efficient document storage and indexing strategies, load balancing, fault tolerance mechanisms, and monitoring automation to ensure system stability under high demand scenarios.

Following the development phases, additional weeks were dedicated to integrated testing, validation with end users, and final deployment preparation. This ensured that the system met both technical quality standards and business expectations before production release.

Change Management and Backlog Prioritization

Change management was handled through a structured backlog prioritization process, aligned with Scrum best practices. Any changes in scope, schedule, or functionality were formally proposed by stakeholders and evaluated based on their impact on the project, expected business value, and required implementation effort. The Product Owner was responsible for prioritizing these changes within the backlog, ensuring that critical requirements were addressed first without compromising project stability.

Sprint review meetings played a key role in this process, allowing stakeholders to evaluate delivered features and provide feedback. Approved changes were incorporated incrementally into subsequent sprints, maintaining agility while ensuring controlled and transparent evolution of the solution.

Integration and Pilot Deployment

The Minimum Viable Product (MVP) was deployed in a cloud-based environment using Google Cloud Platform, leveraging Cloud Run to support a serverless deployment model. This enabled rapid provisioning, scalability, and efficient resource utilization during pilot testing. The deployment process allowed the solution to be validated in a test and near-production environment, ensuring compatibility with the company's infrastructure and operational standards before broader adoption.

Overall, the agile development approach, combined with a structured roadmap and controlled change management process, ensured the successful delivery of an MVP aligned with both technical requirements and business objectives.

2.2.4 **Testing and Technical Evaluation:**

This section presents the testing strategies adopted to validate the proposed solution and the technical evaluation results that demonstrate compliance with the partner company's requirements and quality standards. Given the criticality of accuracy, reliability, and performance in public procurement analysis, a multi-layered testing approach was adopted, covering model evaluation, functional validation, performance assessment, and robustness testing.

Testing Strategy

Testing activities were conducted throughout the development lifecycle, with a particular focus during the final implementation phase of the Minimum Viable Product (MVP). The adopted strategy combined model benchmarking, end-to-end functional testing, user-oriented validation, and performance and robustness tests, reflecting corporate best practices such as User Acceptance Testing (UAT) and system reliability evaluation.

LLM Benchmarking and Model Evaluation

As the core functionality of the system relies on Large Language Models (LLMs) for requirement extraction and document analysis, an objective benchmarking process was conducted to support the model selection decision. Two widely adopted

and comparable models were evaluated under identical conditions: GPT-4.0 mini and DeepSeek V3 0324.

The benchmark focused on a controlled requirement extraction task using a simulated public tender document containing functional requirements, non-functional requirements, project constraints, numerical data, and intentionally introduced linguistic inconsistencies. The evaluation criteria included precision, accuracy, contextual understanding, adherence to output format, response time, and cost efficiency, all of which are critical factors in a corporate environment.

Results indicated that both models successfully extracted the expected functional and non-functional requirements. However, GPT-4.0 mini demonstrated superior precision, better compliance with prompt instructions, and more accurate handling of implicit information, such as identifying misspelled terms and correctly counting numerical references. Although DeepSeek V3 0324 showed advantages in response speed and cost, its limitations in contextual accuracy reinforced the decision to prioritize precision and reliability for the proposed solution. The benchmark also confirmed the necessity of incorporating a vector database to further improve accuracy and reduce hallucinations.

Functional and End-to-End Testing

Functional testing was conducted through end-to-end validation of the entire Retrieval-Augmented Generation (RAG) pipeline, encompassing document ingestion, indexing, semantic retrieval, and answer generation. Real public tender documents in PDF format were used as test inputs to ensure alignment with real-world usage scenarios.

The system successfully processed documents, returned coherent answers to domain-specific queries, and consistently provided traceable sources linked to the original document pages. These tests confirmed that the implemented functionalities

met the defined functional requirements and supported the operational needs of pre-sales users.

Reliability and Consistency Testing

To evaluate system reliability, consistency tests were performed using semantically equivalent questions phrased in different ways. The expected behavior was the generation of consistent answers grounded in the same source documents. The results demonstrated stable behavior, with minimal linguistic variation and consistent source attribution, indicating robustness in the retrieval and generation process.

Performance and Robustness Testing

Performance tests focused on evaluating processing time and response latency under typical usage conditions. Indexing of medium-sized tender documents was completed within acceptable timeframes, and average response times for user queries remained below three seconds, meeting the non-functional performance requirements defined for the system.

Robustness tests were also conducted to assess system behavior under atypical or erroneous inputs, such as irrelevant questions, empty documents, and documents containing complex structures like tables. In all tested scenarios, the system handled errors gracefully, avoided crashes, and returned appropriate fallback responses when information was out of scope or partially unreadable.

Results and Compliance with Technical Requirements

The testing results demonstrate that the proposed solution meets the partner company's technical requirements and quality standards. The system achieved high accuracy in requirement extraction, reliable source traceability, acceptable performance metrics, and robust error handling. These outcomes validate the readiness of the MVP for pilot deployment and confirm its suitability for supporting pre-sales activities in a corporate environment.

Overall, the adopted testing strategy provided strong evidence that the solution is technically sound, reliable, and aligned with both functional and

non-functional requirements, supporting its potential adoption and future scaling within the company.

2.3 Assessment of Impact and Contribution to the Business

This section assesses the impact of the proposed solution from a business perspective, focusing on the return on investment related to time, human resources, and organizational effort required for its implementation. Unlike a traditional financial-only analysis, this assessment considers both tangible and intangible returns associated with process optimization and decision support in pre-sales operations.

From a time investment perspective, the development of the solution required a concentrated effort over a defined project timeline, involving activities such as requirements analysis, model benchmarking, system development, testing, and validation. Although this initial effort represents a non-negligible allocation of technical resources, the results indicate that the investment is justified by the magnitude of the projected operational gains.

Baseline data collected through internal surveys with software architects and the financial team at ATOS indicate that the pre-sales team spends an average of 34 hours analyzing a single public procurement notice. When extrapolated to the annual volume of bids, this represents a significant consumption of senior professionals' time, with high opportunity cost. The proposed solution, even at the MVP level, demonstrates the potential to reduce this effort substantially by automating requirement extraction and structuring, allowing professionals to focus on strategic evaluation rather than manual reading.

From a resource utilization standpoint, the solution contributes to a more efficient allocation of specialized roles. By reducing repetitive and low-value manual tasks, the system enables high-cost professionals to dedicate more time to activities with greater business impact, such as opportunity qualification, risk assessment, and client engagement. This shift directly improves productivity without increasing headcount.

In terms of organizational contribution, the solution introduces a scalable and reusable capability for document analysis that can be extended beyond public tenders to other corporate processes. The knowledge generated throughout the project—including architectural decisions, benchmarking methodologies, prompt engineering practices, and testing strategies—represents an additional return on investment, as it strengthens the company's internal expertise in artificial intelligence and advanced data processing.

Overall, the return on investment of time and resources is evidenced by the balance between a limited, time-bound development effort and the significant, recurring operational benefits projected for pre-sales activities. Even without full production deployment, the solution demonstrates strong potential to deliver sustained business value, justify the initial investment, and support the company's strategic objectives of efficiency, competitiveness, and innovation.

2.3.1 **Defining Corporate Success Metrics:**

Given that the proposed solution was developed as a proof of concept and was not fully deployed in a production environment, the success metrics were

defined based on **business objectives, industry best practices, and validation through controlled tests and simulations**. The KPIs focus on measuring efficiency gains, decision quality, and risk reduction in the pre-sales bid analysis process.

The main KPIs defined for the project are:

1 - Time to Analyze a Bid (Cycle Time)

Measures the average time required to extract and understand technical and financial requirements from a public notice.

- Baseline (Before): Manual analysis conducted by pre-sales specialists.

- Target (After): Automated extraction and summarization using LLM + RAG.

2 - Requirement Extraction Accuracy

Percentage of correctly identified functional and non-functional requirements compared to a reference (expected answer).

- Evaluates precision and hallucination avoidance, which are critical for Go / No-Go decisions.

3 - Decision Support Quality (Go / No-Go Readiness)

Qualitative KPI that assesses whether the generated outputs provide sufficient clarity to support strategic decisions.

- Measured through completeness, clarity, and traceability of extracted requirements.

4 - Response Time per Query

Average latency between user query and system response.

- Important for real-world usability in time-constrained bid processes.

5 - Operational Cost per Analysis

Estimated cost of processing a bid, considering LLM usage, embeddings, and infrastructure.

- Compared against the opportunity cost of senior pre-sales professionals' time.

6 - Source Traceability Rate

Percentage of answers that correctly reference the original document sections or pages.

- Essential for compliance and auditability in public procurement contexts.

2.3.2 **Results and Impact Analysis:**

This section presents the expected results and business impact of the proposed solution, based on baseline metrics collected from the partner company and simulated projections derived from controlled analyses and internal data. As the solution was developed as a Minimum Viable Product (MVP), the results represent **estimated gains** rather than outcomes from a full production deployment.

Baseline Metrics

Based on internal data and a structured survey conducted with pre-sales professionals, the average time required to analyze a single public procurement notice is approximately **34 hours**, considering documents of varying complexity. In 2024, the pre-sales team handled around **50 bidding processes**, resulting in an estimated **306 hours per month** dedicated exclusively to bid analysis activities.

From a financial perspective, the total operational cost associated with bid analysis reached approximately **R$ 52,538.76 per month**, considering the involvement of senior roles such as Solution Manager, Financial Architect, Bid Manager, and CDE specialists.

Quantitative Impact (Projected Results)

With the adoption of the proposed AI-based solution, the estimated effort required to analyze a single notice is significantly reduced. Projections indicate a reduction to approximately **72 hours per month** for the same average volume of bids, representing a **reduction of nearly 76% in analysis time**.

This efficiency gain leads to a projected reduction in monthly operational costs from **R$ 52,538.76** to approximately **R$ 17,492.58**, resulting in an estimated **monthly savings of R$ 35,000**. Based on the initial development investment and recurring maintenance costs, the projected **payback period is approximately seven months**, indicating strong financial viability.

Qualitative Impact

In addition to quantitative gains, the solution is expected to generate significant qualitative benefits. These include increased agility in Go / No-Go decision-making, improved consistency and reliability in requirement analysis, better allocation of senior professionals to higher-value activities, and enhanced

competitiveness in public procurement processes. The availability of structured, traceable, and summarized information also contributes to improved decision quality and reduced operational risk.

Data Source and Measurement Methodology

The baseline metrics and financial estimates presented in this analysis were derived from **internal data collection activities conducted with the partner company**. Quantitative and qualitative inputs were gathered through structured surveys and interviews with **software architects from the pre-sales technical team** and professionals from the **financial department of ATOS**.

The collected data included average time spent on bid analysis, team composition, cost per hour of specialized roles, and overall operational effort associated with public procurement processes. These inputs were used to establish realistic baseline metrics and to simulate the projected impact of the proposed AI-based solution under controlled conditions.

As the system was developed as a Minimum Viable Product (MVP) and was not deployed in a production environment, the “after” metrics represent **projected results**, based on experimental tests, controlled benchmarks, and validated assumptions aligned with real operational workflows. This approach ensures methodological rigor while maintaining transparency regarding the nature of the results.

2.3.3 **Cost-Benefit Analysis:**

The cost-benefit analysis was conducted based on estimated development, implementation, and maintenance costs, combined with projected operational savings resulting from process optimization.

Estimated Costs

The initial investment for solution development, including multidisciplinary team allocation and system setup, was estimated at approximately **R$ 248,415**, incorporating a safety margin. Monthly operational and maintenance costs, covering infrastructure, cloud services, and a reduced support team, were estimated at approximately **R$ 8,124**, also including a contingency margin.

Return on Investment (ROI) and Payback

When compared to the baseline operational cost of bid analysis, the proposed solution demonstrates a strong economic justification. With projected monthly savings of approximately **R$ 35,000**, the estimated payback period is around **seven months**. Beyond this point, the solution transitions into a cost-saving asset for the organization, while also enabling scalability and process standardization.

In addition to direct financial returns, the investment contributes to strategic benefits, such as increased competitiveness in public tenders, higher bid conversion potential, and long-term modernization of internal processes through artificial intelligence.

2.3.4 **Critical Success Factors and Lessons Learned:**

Several factors contributed positively to the development and potential impact of the proposed solution, even though it was not fully deployed in a production environment.

1. **Clear Business Alignment with Pre-Sales Needs**

The solution was designed with a strong focus on improving internal processes within pre-sales teams, particularly in the analysis of public procurement notices. By addressing a real operational pain point—high time consumption and opportunity cost—the project maintained clear relevance to the company's strategic objectives of efficiency and competitiveness.

2. **Use of Emerging AI Technologies with Market Validation**

The adoption of Large Language Models (LLMs), vector databases, and Retrieval-Augmented Generation (RAG) reflects current market best practices for document-intensive processes. Benchmarking different models under controlled conditions ensured informed technical decisions and reduced the risk of adopting inadequate technologies.

3. **Focus on Process Optimization Rather Than Full Automation**

The solution was positioned as a decision-support tool, not a replacement for human expertise. This approach is critical in regulated environments such as public procurement, where compliance, interpretability, and auditability are essential.

4. **Incremental and Value-Oriented Development Approach**

The project was developed iteratively, with continuous validation of outputs at each stage. This ensured that each increment delivered tangible value to the partner company, even at a proof-of-concept level.

Limiting Factors and Challenges

Despite the positive outcomes, some factors limited the immediate impact of the solution:

1. **Absence of Production Deployment**

The lack of real-world deployment prevented the measurement of long-term business KPIs, such as reduction in Go / No-Go decision errors or improvement in bid success rates.

2. **Document Variability and Complexity**

Public notices vary significantly in structure, formatting, and terminology, which can affect extraction accuracy. This reinforced the need for more advanced chunking and document segmentation strategies.

3. **Dependence on Prompt and Context Design**

The quality of LLM outputs proved to be highly sensitive to prompt engineering and context structuring, requiring careful tuning to avoid hallucinations and incomplete answers.

Lessons Learned

- Artificial Intelligence can significantly enhance internal processes when applied with a **clear business objective**, rather than as a purely experimental technology.

- Combining LLMs with vector databases is essential to achieve acceptable accuracy and traceability in complex, domain-specific documents.

- Early benchmarking and testing are critical to avoid premature technological commitments and to balance accuracy, performance, and cost.

- Even limited-scope prototypes can generate actionable insights and guide strategic decision-making within the company.

Next Steps and Future Evolution

Based on the results and lessons learned, the following steps are recommended to increase the solution's business impact:

1. **Expansion of Analysis Scope**

Extend the solution to support additional document types, such as contracts, annexes, technical specifications, and compliance reports, enabling a more comprehensive bid evaluation process.

2. **Improved Chunking Strategy**

Implement document chunking by logical sections (e.g., technical requirements, financial terms, compliance clauses) to improve retrieval accuracy and contextual relevance of responses.

3. **Automated Match Scoring Between Notices and Technical Certificates**

Introduce a percentage-based matching mechanism to quantify the alignment between bid requirements and the company's technical certifications (atestados), supporting faster and more objective Go / No-Go decisions.

4. **Integration with Corporate Systems**

In future iterations, the solution could be integrated with internal document repositories and CRM or bid management tools, increasing adoption and operational efficiency.

3 Conclusion

This work presented the development of an integrated solution based on Large Language Models (LLMs) aimed at supporting the analysis of public procurement tenders within a consulting company operating in the public sector. The primary objective of the project was to investigate how artificial intelligence could be applied to optimize internal pre-sales processes, specifically by reducing the time and effort required to interpret extensive bid documents and by improving the identification of technical and financial requirements relevant to Go / No-Go decision-making.

Based on the results obtained throughout the project, it can be concluded that the proposed objectives were achieved. The developed solution demonstrated technical feasibility and strategic relevance by enabling automated extraction, structuring, and contextual analysis of tender documents through the combined use of LLMs, vector databases, and Retrieval-Augmented Generation (RAG). Controlled tests, model benchmarking, and end-to-end validation confirmed that the system is capable of producing accurate, traceable, and consistent outputs, aligned with the needs of pre-sales teams operating under strict deadlines and regulatory constraints.

From a business perspective, the project highlights the significant potential impact of adopting artificial intelligence to optimize document-intensive processes. Baseline metrics collected through internal surveys with software architects and the financial team at ATOS revealed a high operational cost and substantial opportunity cost associated with manual bid analysis. The projected results indicate that the solution could lead to substantial reductions in analysis time and operational expenses, while simultaneously increasing agility, consistency, and decision quality.

Even as a Minimum Viable Product (MVP), the solution provides strong evidence of its value as a decision-support tool capable of enhancing competitiveness in the public procurement market.

In addition to quantitative benefits, the project delivered important qualitative gains. The availability of structured and summarized information improves transparency and confidence in decision-making, reduces reliance on individual expertise, and enables better allocation of senior professionals to higher-value strategic activities. These aspects are particularly relevant in the context of pre-sales operations, where errors or delays in early-stage analysis can directly impact revenue generation and market positioning.

Based on the lessons learned and the results obtained, several recommendations can be made regarding the future evolution of the solution. First, the scope of document analysis can be expanded to include additional document types, such as contracts, annexes, technical specifications, and compliance reports, enabling a more comprehensive evaluation of bidding opportunities. Second, improvements in document chunking strategies—such as segmentation by logical sections—can further enhance retrieval accuracy and contextual relevance. Third, the implementation of a quantitative matching mechanism, expressed as a percentage of alignment between tender requirements and corporate certifications, would provide a more objective basis for Go / No-Go decisions. Finally, scaling the solution through deeper integration with corporate systems, such as document repositories, bid management tools, and CRM platforms, would increase adoption and operational impact.

Regarding scalability and maintenance, the cloud-native architecture adopted in this project provides a solid foundation for future growth. The use of managed services and serverless infrastructure facilitates scalability, cost control, and operational resilience. To ensure long-term sustainability, it is recommended that the solution be accompanied by comprehensive technical documentation, including architectural diagrams, API specifications, prompt design guidelines, and operational procedures. This documentation is essential to support knowledge transfer and enable the company’s internal technical teams to maintain, evolve, and extend the solution independently.

A structured knowledge transfer plan is also recommended as a next step. This plan should include technical walkthrough sessions, documentation repositories, and pilot usage with selected pre-sales teams, ensuring that both technical and business stakeholders fully understand the system's capabilities and limitations. Such an approach would reduce dependency on external development and accelerate internal adoption.

Finally, it is important to acknowledge the institutional and human contributions that made this project possible. The author would like to express sincere gratitude to ATOS for providing the corporate environment, technical context, and access to domain expertise that enabled the development of a solution aligned with real business challenges. Special thanks are extended to the pre-sales teams, software architects, and financial professionals who contributed insights, data, and feedback throughout the project. The author also wishes to thank the academic advisor for the guidance, availability, and valuable contributions provided since the early stages of the work, which were essential for structuring the project and ensuring its academic rigor.

In conclusion, this project demonstrates that the strategic application of artificial intelligence can play a decisive role in improving internal processes, supporting decision-making, and increasing competitiveness in consulting companies operating in the public sector. Beyond its immediate results, the work lays a solid foundation for future initiatives that leverage AI as a catalyst for organizational efficiency and innovation.

References

ABUOBIEDA, A.; SALIM, N.; KUMAR, Y. J.; OSMAN, A. H.

An improved evolutionary algorithm for extractive text summarization.

In: **INTELLIGENT INFORMATION AND DATABASE SYSTEMS**. Berlin: Springer, 2013. p. 78–89.

ABUOBIEDA, A.; SALIM, N.; KUMAR, Y. J.; OSMAN, A. H.

Opposition differential evolution based method for text summarization.

In: **INTELLIGENT INFORMATION AND DATABASE SYSTEMS**. Berlin: Springer, 2013. p. 487–496.

ALLIHEEDI, M.; DI MARCO, C.

Rhetorical figuration as a metric in text summarization.

In: **ADVANCES IN ARTIFICIAL INTELLIGENCE**. Berlin: Springer, 2014. p. 13–22.

BATCHA, N. K.; AZIZ, N. A.; SHAFIE, S. I.

CRF-based feature extraction applied for supervised automatic text summarization.

Procedia Technology, Amsterdam, v. 11, p. 426–436, 2013.